

RAJARATA UNIVERSITY OF SRI LANKA, MIHINTALE
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCES

FDN 1204/ FDN 1306– Basic Mathematics
Tutorial

01) Show that the square matrix $A = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 5 & 3 \\ -1 & 8 & 2 \end{pmatrix}$ is a singular matrix.

02) If $A = \begin{pmatrix} 1 & 4 & 3 \\ 6 & 2 & 5 \\ 1 & 7 & 0 \end{pmatrix}$, Determine
i. $|A|$ ii. $\text{Adj } A$

03) Find the inverse of the matrix $A = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 5 & 1 \\ 2 & 0 & 6 \end{pmatrix}$

04) If $A = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ determine

i. B^{-1} ii. AB iii. $B^{-1}A$

05) Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 5 & 7 \end{pmatrix}$

i. Compute $|A|$ iii. Verify $A(\text{adj } A) = |A|I$
ii. Find $\text{adj } A$ iv. Find A^{-1}

06) Find the possible value of x can take, Given that

$$A = \begin{pmatrix} x^2 & 3 \\ 1 & 3x \end{pmatrix} \quad B = \begin{pmatrix} 3 & 6 \\ 2 & x \end{pmatrix} \text{ Such that } AB = BA$$

07) If $A = \begin{pmatrix} 3 & 2 \\ -4 & 1 \end{pmatrix}$ Find the values of m and n given that $A^2 = mA + nA$

08) If matrix A is $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{pmatrix}$ and I , the unit matrix of order 3, Show that $A^3 = pI + qA + rA^2$

09) Find a matrix K such that $AKB = C$ given that

$$A = \begin{pmatrix} 1 & 4 \\ -2 & 3 \\ 1 & -2 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} \quad C = \begin{pmatrix} 8 & 6 & -6 \\ 6 & -1 & 1 \\ -4 & 0 & 0 \end{pmatrix}$$